

SAHANA ARULANANDAM

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SUMMARY

I am an enthusiastic and creative robotics and mechatronics engineering student, currently pursuing my degree at Christ University, Bengaluru. I am driven by a passion for autonomous systems, robot simulation, and intelligent IoT applications. With hands-on experience in ROS 2 navigation, AMR simulation, swarm robotics, and digital twin development, I excel at tackling cross-disciplinary challenges. As a team leader and researcher, I bring strong technical depth across embedded systems, computer vision, and software development. I am eager to join an innovative, collaborative team where I can leverage my technical expertise and creative problem-solving skills to advance robotics and automation technologies, contributing to projects that will help redefine the future of intelligent autonomous systems and human-robot interaction.

TECHNICAL SKILLS

Robotics & Simulation

- **Frameworks:** ROS 2 (Jazzy), Gazebo Harmonic
- **Navigation:** Differential Drive & AMR Navigation, Path Planning, Obstacle Avoidance, LiDAR, SLAM, PID Control, PWM Motor control, URDFs
- **Sensing:** Sensor Integration (LiDAR, IMU, Ultrasonic, Encoders)
- **3D Modelling:** Blender, Solidworks

IoT & Hardware

- **Microcontrollers:** Raspberry Pi, Nvidia Jetson Orin Nano, ESP8266 / ESP32, Arduino Uno / Mega / Nano
- **Protocols:** MQTT, UDP, TCP, ESP-NOW, CoAP Protocols, ThingSpeak Cloud Integration, Blynk IoT, Eclipse Mosquitto

Programming:

- Python, C++, C, Java, Linux, Ubuntu, Git, MATLAB & Simulink

Languages:

- English, Hindi, Tamil

EDUCATION

- **Christ University, Bengaluru 2023 - 2027**
B.Tech in Robotics and Mechatronics Engineering
- **Presentation Convent School, Nerul 2019 - 2023**
ICSE & ISC Classes 10th and 12th

PUBLICATIONS

- **Conference Paper** "Comparative Evaluation of DWB, MPPI, and RPP Local Planners for Autonomous Mobile Robots in Indoor Environments" *DSATM AMISE 2026*
- **Conference Paper** "Design and Implementation of Hand-Gesture Controlled Robotic Arm using Smart Gloves and Machine Learning" *DSI-DMMES 2026*
- **Conference Poster** "Hybrid Communication and Digital Twin Integration for Autonomous Mobility" *ICVTTS 2025*

WORK EXPERIENCE

Intern April 2023 – May 2023

ChattingBirds Pvt Ltd, Hosur, India

- Built foundational knowledge of AMR architectures, robotics mathematics, and HMI design patterns through hands-on simulation using ROS 2.
- Studied and modeled navigation, path planning, sensor communication systems, and real-time data flow for industrial and logistics environments.

Undergraduate Research Intern April 2024 – June 2024

Christ (Deemed To Be University), Centre of Excellence in E-Mobility, Bengaluru, India

- Contributed to a digital twin of an electric two-wheeler in Unreal Engine 5, integrating ESP8266 sensors and UDP for real-time data transmission.
- Enabled cloud-based monitoring of electrical parameters via ThingSpeak with synchronized physical-to-virtual sensor representation.

PROJECTS

Autonomous Mobile Robot (AMR) Navigation System

- Implemented a ROS 2 Jazzy and Gazebo Harmonic on Nvidia Jetson Orin Nano-based robot for real-time localization and path planning in simulated environments.
- Evaluated and compared DWB, MPPI, and RPP local planners on both simulation and hardware to optimize navigation performance and obstacle avoidance behaviour.

Collaborative Swarm Robotics System

- Designed a multi-agent coordination framework using MQTT for decentralized communication and dynamic task allocation among autonomous robots.
- Implemented PID-based collision avoidance logic to maintain safe inter-robot spacing and ensure reliable swarm coordination.

Autonomous Guided Robot (AGV)

- Led the programming team for a line-following AGV using an IR sensor array interfaced with an Arduino Mega and Raspberry Pi 5 for reliable autonomous path tracking.
- Implemented safety features including speed limits, emergency stop, and manual override to ensure safe and controlled operation.

LEADERSHIP ROLES

Founding Chair - IEEE Robotics and Automation Society (RAS), Christ University | 2025 – 2026

- Led the student branch chapter of the IEEE RAS, fostering a community of robotics and automation enthusiasts on campus.
- Organizing technical events, workshops, and initiatives to promote robotics education and industry exposure among students.

Department Head - Magnovite 2025, Christ University | December 2025

- Headed a department at Magnovite, Christ University's flagship annual fest, overseeing planning, coordination, and execution of departmental events.
- Managed a team across logistics, promotions, and on-ground operations to ensure smooth delivery of fest activities.

CERTIFICATIONS

NPTEL Swayam - Advanced Robotics Applications

MathWorks - MATLAB & Simulink Onramp, MATLAB & Simulink Fundamentals